BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 108 · Lesson 108 · Interrupts and the Interrupt Service Routine

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. With page size 1,024 bytes, what offset is address 2,500?

Q2. For the binary value 1010, what is its decimal value?

Q3. A linked-list node has data and what other essential field?

Q4. In a BST, where is a value less than the current node placed?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

interrupt — Signal requiring CPU attention.

ISR — Routine that services a particular interrupt.

context — CPU state needed to resume execution.

polling — Repeated checking of a device or condition.

New learning

Event occurs — A device or timer signals that attention is needed.

State is preserved — The CPU completes a suitable instruction boundary and saves essential context.

Service and return — The ISR handles the event; saved context is restored and execution resumes.

Worked example

A keyboard interrupt arrives while a calculation is running. Trace the CPU response.

1. Model answer — After completing the current instruction and recognising the interrupt, the CPU preserves relevant state, runs the keyboard ISR to obtain the key event, restores state and resumes the interrupted program. — Check: link each claim to the scenario and use precise OCR terminology.

We Do · Practise with guidance

A disk controller signals that reading a sector has finished. Explain the sequence.

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

The disk raises an interrupt; the CPU saves context when accepted; the ISR acknowledges completion and updates status; the original process resumes or the scheduler selects another ready process.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Outline the role of an interrupt and an ISR. [6]

Question 2

Explain where an interrupt can fit into the fetch–decode–execute cycle. [6]

Question 3

Evaluate interrupts compared with repeatedly polling a low-frequency sensor. [8]

Exit Ticket · Answer before checking

An ISR should normally preserve enough execution state to allow the interrupted work to resume.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 452.

Q2. 10.

Q3. A link/reference to the next node (for a singly linked list).

Q4. In its left subtree (under the chosen no-duplicate convention).

You Do pages · Independent questions

Q1. Interrupt signals attention [2]; ISR is event-specific handler [2]; context is saved/restored to resume processing [2].

Q2. Normally checked at a suitable boundary, often after an instruction [2]; current state preserved [2]; ISR executed before next normal instruction and context restored [2].

Q3. Explain polling [2], event-driven interrupt [2], reduced wasted CPU checking [2], and interrupt overhead/priority or timing trade-off [2].

Exit Ticket · Model answer

True. Relevant context is saved and restored.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.